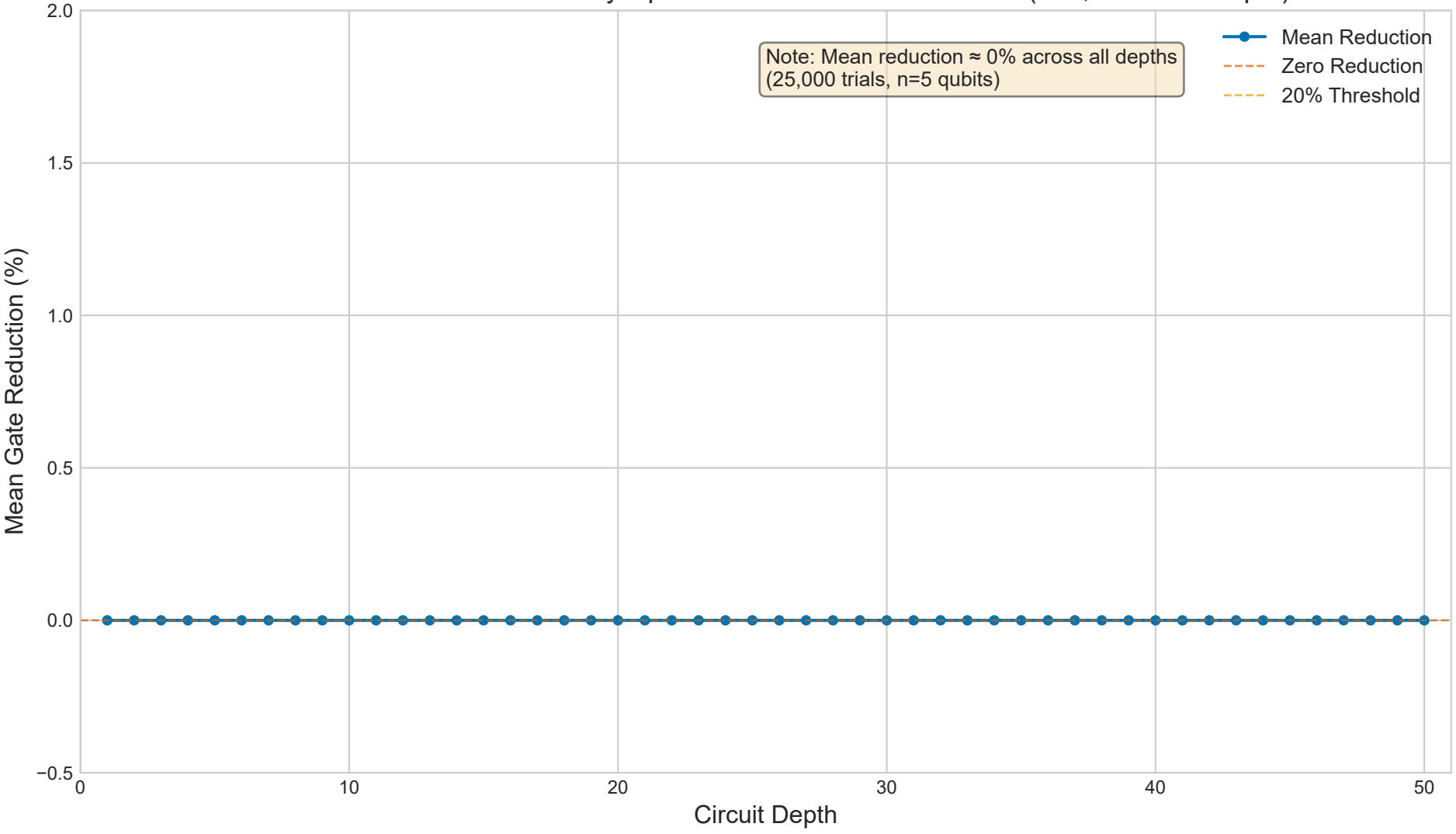
E1: Phase Transition — Greedy Optimization on Random Circuits (n=5, 500 trials/depth)
Note: Mean reduction ≈ 0% across all depths
(25,000 trials, n=5 qubits)
Mean Reduction
Zero Reduction
20% Threshold
Mean Gate Reduction (%)
Circuit Depth
2.0
1.5
1.0
0.5
0.0
−0.5
0
10
20
30
40
50